

Full Metadata Document: Platform Source Simulator

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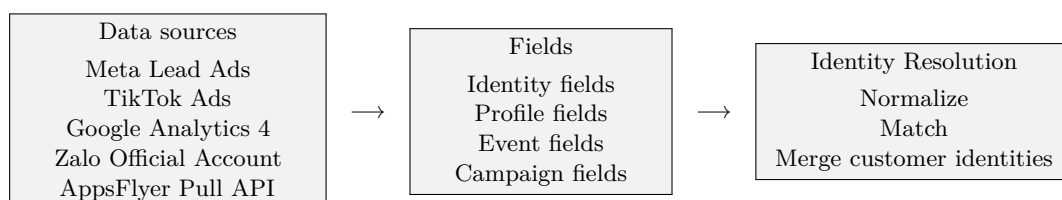
Contents

Abstract	1
1. Document Scope	2
2. Shared Identity Metadata	2
3. Meta Lead Ads Metadata	4
4. TikTok Integrated Reports Metadata	6
5. Google Analytics 4 Metadata	8
6. Zalo Official Account Metadata	12
7. AppsFlyer Pull API Raw-Data Metadata	14
8. Cross-Platform CIR Metadata	16
9. Source References and Maintenance Rules	16
10. Pandoc PDF Export	17

Abstract

This document is the metadata contract for `all-data-simulator/full_raw_data_simulator.py`. It describes the source platform represented by each dataset, the generated CSV schema, field types and meanings, identity value, and differences between a real platform payload and this simulator's flattened test representation.

Overview Flow



1. Document Scope

Source datasets

Dataset	Source platform	Grain
meta_cir_api.csv	Meta Lead Ads	One lead per row
tiktok_cir_api.csv	TikTok Ads	One ad/reporting-day row
ga4_cir_api.csv	Google Analytics 4	One synthetic event/report row
zalo_cir_api.csv	Zalo Official Account	One user profile row
appsflyer_cir_api.csv	AppsFlyer	One in-app event row

Generator index

Generator	Dataset
generate_meta_data	meta_cir_api.csv
generate_tiktok_data	tiktok_cir_api.csv
generate_ga4_data	ga4_cir_api.csv
generate_zalo_data	zalo_cir_api.csv
generate_appsflyer_data	appsflyer_cir_api.csv

Runtime and output metadata

Item	Value
Default row count	20 rows per dataset
Shared identity count	8 people, reused cyclically
Base date/time	2026-08-01 09:00:00
Random seed	42
CSV encoding	utf-8-sig (UTF-8 with BOM)
CSV newline mode	newline=""
Output directory	all-data-simulator/platform_cir_csv/
ZIP archive	platform_cir_api_csv_simulated.zip under all-data-simulator/
ZIP contents	The five generated CSV files, stored by basename
Missing dictionary keys	Ignored by <code>csv.DictWriter</code> because <code>extrasaction="ignore"</code>

Relative output paths are resolved from the process working directory, not from the script directory. The documented command assumes the repository root is the working directory.

2. Shared Identity Metadata

PEOPLE is the synthetic identity source used to make cross-platform identity resolution possible. For row index i , each generator selects `PEOPLE[i % len(PEOPLE)]` except for TikTok, which is aggregate advertising data and does not use a person record.

Attribute	Type	Example	Purpose
cid	string	CUST-0001	Synthetic customer key; appears in AppsFlyer <code>customer_user_id</code> and JSON <code>event_value</code>
name	string	Nguyen An	Full display name for Meta and Zalo
first	string	An	Meta lead first name
last	string	Nguyen	Meta lead last name

Attribute	Type	Example	Purpose
email	string	an.nguyen@example.test	Meta lead email; synthetic and non-deliverable
phone	string	84901234001	Meta/Zalo phone; digits are intentionally normalized-looking
city	string	Hanoi	Geographic identity/context field
state	string	Hanoi	Geographic identity/context field
country	string	VN	ISO-like country code used in platform rows
gender	integer	1	Zalo user gender simulation: code comments define 1 male, 2 female, 0 other/unknown

Identity linkage is synthetic. Real platform IDs are not used as a shared customer key. The strongest simulated CIR keys are email, phone, name, and the AppsFlyer `customer_user_id/event_value.customer_id` pair.

3. Meta Lead Ads Metadata

Source contract

- Official guide: [Meta Lead Ads retrieval guide](#)
- Lead node: [Meta Lead object reference](#)
- Lead form: [Meta Leadgen Form reference](#)

A real Meta lead is an object. Form answers are commonly represented in a nested `field_data` collection. This simulator emits a flat CSV projection for ingestion and CIR testing.

Dataset metadata

Item	Value
File	<code>meta_cir_api.csv</code>
Grain	One synthetic lead
Rows	<code>Config.NUM_ROWS</code>
Primary source identifier	<code>id</code>
Identity fields	<code>field_email</code> , <code>field_phone</code> , <code>field_first_name</code> , <code>field_last_name</code> , <code>field_city</code> , <code>field_state</code> , <code>field_country</code>
Time field	<code>created_time</code>
Time format	<code>YYYY-MM-DDTHH:MM:SS+0000</code>
Flattening	Form answers become <code>field_*</code> columns; campaign hierarchy is also flattened

Column metadata

Column	Type	Nullability	Meaning and generation
<code>id</code>	string	non-null	Synthetic Meta lead object ID, formatted as <code>3280000000NNNN</code> ; simulator primary key
<code>created_time</code>	string datetime	non-null	<code>Config.BASE_DATE + i*4 hours + random minute</code> , formatted with +0000 offset
<code>form_id</code>	string	non-null	Fixed synthetic lead form ID <code>890000123456789</code>
<code>campaign_id</code>	string	non-null	Synthetic campaign ID; random suffix from 1 to 4, so values can repeat
<code>campaign_name</code>	string	non-null	Synthetic campaign label; independently randomized from 1 to 4
<code>adset_id</code>	string	non-null	Synthetic ad set ID; random suffix from 1 to 5
<code>adset_name</code>	string	non-null	Fixed synthetic ad set name <code>Broad_Audience_18_65</code>
<code>ad_id</code>	string	non-null	Synthetic ad ID; random suffix from 1 to 7
<code>ad_name</code>	string	non-null	Synthetic ad name; random creative suffix from 1 to 7
<code>field_email</code>	string email	non-null	Email answer from the shared synthetic person; strongest Meta CIR key
<code>field_phone</code>	string phone	non-null	Phone answer from the shared synthetic person; strongest Meta CIR key after normalization

Column	Type	Nullability	Meaning and generation
<code>field_first_name</code>	string	non-null	First-name form answer
<code>field_last_name</code>	string	non-null	Last-name form answer
<code>field_city</code>	string	non-null	City form answer
<code>field_state</code>	string	non-null	State/province form answer
<code>field_country</code>	string ISO-like code	non-null	Country form answer, currently VN

Simulator limitations

- `field_*` names are simulator column names, not a literal copy of a Meta response object's nested `field_data` structure.
- Campaign, ad set, and ad IDs are not guaranteed to be unique or stable by entity.
- No Meta API permissions, access tokens, lead retrieval pagination, or webhook fields are represented.

4. TikTok Integrated Reports Metadata

Source contract

- API portal overview: [TikTok Marketing API reporting overview](#)
- Integrated report endpoint family: `/open_api/v1.3/report/integrated/get/`

TikTok reports contain dimensions and metrics. The simulator models an ad-level basic report grouped by day and country. The real API commonly returns numeric response values as strings in JSON; this CSV simulator writes Python numeric values, which the CSV serializer renders without quotes.

Dataset metadata

Item	Value
File	<code>tiktok_cir_api.csv</code>
Grain	One synthetic ad/reporting-day row
Rows	<code>Config.NUM_ROWS</code>
Dimensions	Advertiser, date, campaign, ad group, ad, country
Metrics	Impressions, clicks, spend, CTR, CPC, CPM, conversions, conversion rate
Reporting date	<code>stat_time_day</code>
Reporting date format	<code>YYYY-MM-DD 00:00:00</code>
Currency	Not explicitly declared; <code>spend</code> is a synthetic account-currency amount

Column metadata

Column	Type	Nullability	Meaning and generation
<code>advertiser_id</code>	string	non-null	Fixed synthetic advertiser ID 710000123456789
<code>stat_time_day</code>	string datetime	non-null	<code>Config.BASE_DATE + (i % 10) days</code> , normalized to midnight
<code>campaign_id</code>	string	non-null	Synthetic campaign ID; random suffix from 1 to 4
<code>campaign_name</code>	string	non-null	Synthetic campaign name; independently randomized from 1 to 4
<code>adgroup_id</code>	string	non-null	Synthetic ad group ID; random suffix from 1 to 5
<code>adgroup_name</code>	string	non-null	Synthetic ad group name; random suffix from 1 to 5
<code>ad_id</code>	string	non-null	Synthetic ad ID; random suffix from 1 to 7
<code>ad_name</code>	string	non-null	Synthetic ad name; random suffix from 1 to 7
<code>country_code</code>	string ISO-like code	non-null	Random choice from VN, SG, and TH; VN has four entries and is therefore weighted
<code>impressions</code>	integer	non-null	Random integer from 10,000 to 20,000
<code>clicks</code>	integer	non-null	Integer truncation of impressions multiplied by a random rate from 1% to 4%
<code>spend</code>	decimal	non-null	Rounded to two decimals; clicks multiplied by a random amount from 0.3 to 0.8

Column	Type	Nullability	Meaning and generation
ctr	decimal percent	non-null	clicks / impressions * 100 , rounded to four decimals
cpc	decimal	non-null	spend / clicks , rounded to four decimals; zero if clicks are zero
cpm	decimal	non-null	spend / impressions * 1000 , rounded to four decimals
conversion	integer	non-null	Integer truncation of clicks multiplied by a random rate from 1% to 5%
conversion_rate	decimal percent	non-null	conversion / clicks * 100 , rounded to four decimals; zero if clicks are zero

Simulator limitations

- The report request, access token, pagination, account timezone, account currency, and API response envelope are not represented.
- Dimension IDs and names are independently randomized, so a repeated ID may receive different generated names across rows.
- Metrics are synthetic estimates and should not be treated as platform benchmarks.

5. Google Analytics 4 Metadata

Source contract

- Official Data API schema: [Google Analytics Data API dimensions and metrics](#)
- GA4 configuration fields: [Google Analytics configuration reference](#)
- Send login identity: [Send user IDs](#)
- Send hashed user data: [Send user-provided data using Measurement Protocol](#)
- Measurement Protocol payload: [Measurement Protocol reference](#)
- PII policy: [Best practices to avoid sending PII](#)

This dataset uses Google Analytics Data API dimension and metric names. It is a flat CSV projection of synthetic report rows, not a GA4 BigQuery export event record.

GA4 attribution and identity integration

UTM source and Facebook traffic For a website, GA4 can read campaign parameters from the landing URL, including `utm_source`, `utm_medium`, `utm_campaign`, `utm_id`, `utm_content`, and `utm_term`. These values feed GA4 traffic-source dimensions such as `sessionSource`, `sessionMedium`, `sessionCampaignName`, and `sessionCampaignId`.

`utm_source=facebook` identifies Facebook as the traffic source; it does not identify the customer. Meta's `fbclid` is a click identifier, not a customer ID, `cid`, or authenticated user identity. It must not be used as a CIR customer key. If the application retains it for internal campaign diagnostics, do not place PII in campaign parameters or page URLs, and apply the project's privacy and retention rules.

Do not confuse these identifiers: GA4 `client_id` identifies a browser or app instance, GA4 `user_id` identifies a signed-in user with a non-PII site-owned value, `fbclid` identifies a Meta ad click, and the simulator's `cid` identifies a synthetic customer.

The simulator now generates synthetic UTM fields and a Facebook-only `fbclid` as collection metadata. These values represent landing-page attribution inputs; they are not copied from a real URL or Facebook account.

Google SSO or internal login After a successful Google SSO or internal login, the website may set GA4 `user_id` to a stable, site-owned, pseudonymous identifier. Google documents `user_id` as a reserved configuration value for connecting activity across sessions, devices, and platforms. The value must not itself be PII. It must not be sent as a custom user property, a custom dimension, or an event-level parameter.

Send `user_id` only while the user is authenticated. On logout, set it to `null`; do not send an empty string or the text `"null"`. The simulator represents anonymous rows with empty `user_id` and authenticated rows with a synthetic non-PII value. It also includes synthetic `client_id`, `user_pseudo_id`, and `login_method` collection fields in the CSV.

Hashed email and phone When a server-side integration uses GA4 Measurement Protocol, user-provided data can be sent in the top-level `user_data` object alongside `user_id`. Email and phone are not ordinary event parameters in this model. Measurement Protocol requires the developer to normalize the values, hash them with SHA-256, and send lowercase hex-encoded values in `sha256_email_address` and `sha256_phone_number`.

The documented normalization rules include:

- Email: remove leading/trailing whitespace, lowercase, remove spaces, and remove periods before the domain for `gmail.com` or `googlemail.com` addresses, then hash.
- Phone: remove all non-digit characters, add the `+` prefix using E.164-style form, then hash.

Only send this data when the required consent, privacy, and data-governance conditions for the implementation are satisfied. Never send raw email or phone in a GA4 URL, ordinary event parameter, or custom dimension. The simulator CSV contains only synthetic SHA-256 fixture values in the `sha256_*` columns; it does not contain raw email or phone.

Illustrative Measurement Protocol shape:

```
{
  "client_id": "WEB_CLIENT_ID",
  "user_id": "NON_PII_INTERNAL_USER_ID",
  "events": [{"name": "login", "params": {"method": "google_sso"}}],
  "user_data": {
    "sha256_email_address": ["SHA256_EMAIL_HEX"],
    "sha256_phone_number": ["SHA256_PHONE_HEX"]
  }
}
```


The example values are placeholders. The actual request also requires the appropriate Measurement Protocol transport credentials and identifiers.

Dataset metadata

Item	Value
File	<code>ga4_cir_api.csv</code>
Grain	One synthetic event/report row
Rows	<code>Config.NUM_ROWS</code>
Date field	<code>date</code>
Date format	<code>YYYYMMDD</code>
Event set	<code>page_view</code> , <code>view_item</code> , <code>begin_checkout</code> , <code>purchase</code>
Source set	<code>google</code> , <code>facebook</code> , <code>tiktok</code> , <code>zalo</code>
Revenue currency	Not explicitly declared; values are synthetic VND-like amounts
Key-event model	<code>purchase</code> produces <code>keyEvents = 1</code> ; all other events produce 0
Authentication model	Every third row is anonymous; other rows use synthetic <code>google_sso</code> or <code>internal</code> login context
UTM model	<code>utm_*</code> values are synthetic and aligned with the session/campaign fields
User data model	Authenticated rows contain synthetic SHA-256 email/phone values; raw PII is never generated

Column metadata

Column	Type	Nullability	Meaning and generation
<code>date</code>	string date	non-null	Random date from base date through 10 days later, formatted <code>YYYYMMDD</code>
<code>eventName</code>	string enum-like	non-null	Random synthetic event from the four-event set
<code>campaignId</code>	string	non-null	Synthetic <code>GAD-C360-NNN</code> ID, random suffix from 1 to 4
<code>campaignName</code>	string	non-null	Synthetic web campaign name, independently randomized from 1 to 4
<code>sessionSource</code>	string enum-like	non-null	Random choice from <code>google</code> , <code>facebook</code> , <code>tiktok</code> , <code>zalo</code>
<code>sessionMedium</code>	string enum-like	non-null	Intended to be <code>cpc</code> for Google and <code>paid_social</code> otherwise; current implementation compares the source list rather than the selected source, so it currently emits <code>paid_social</code> for every row
<code>country</code>	string ISO-like code	non-null	Shared synthetic person's country, currently <code>VN</code>
<code>city</code>	string	non-null	Shared synthetic person's city
<code>browser</code>	string enum-like	non-null	Random choice from <code>Chrome</code> , <code>Safari</code> , <code>Edge</code>
<code>deviceCategory</code>	string enum-like	non-null	Random choice from <code>mobile</code> , <code>desktop</code> , <code>tablet</code>
<code>platform</code>	string enum-like	non-null	Fixed <code>web</code>

Column	Type	Nullability	Meaning and generation
<code>sessionCampaignId</code>	string	non-null	Same generated value as <code>campaignId</code>
<code>sessionCampaignName</code>	string	non-null	Synthetic session campaign name; independently randomized from 1 to 4
<code>transactionId</code>	string	empty for non-purchase	<code>ORD-NNNN</code> for purchase rows; empty string otherwise
<code>eventCount</code>	integer	non-null	Random integer from 1 to 5
<code>keyEvents</code>	integer	non-null	1 for purchase rows and 0 otherwise; modern GA4 metric naming
<code>totalRevenue</code>	decimal	non-null	Random 450,000 to 950,000 for purchase rows; 0 otherwise
<code>engagementRate</code>	decimal fraction	non-null	Random fraction from 0.35 to 0.85; follows GA4 fraction convention
<code>client_id</code>	string	non-null	Synthetic browser/app collection identifier, unique per generated row; not a customer ID
<code>user_pseudo_id</code>	string	non-null	Synthetic pseudonymous GA4 collection identifier, unique per generated row
<code>user_id</code>	string	empty for anonymous rows	Synthetic non-PII site-owned identifier <code>GA4-CUST-NNNN</code> for authenticated rows
<code>login_method</code>	string	empty for anonymous rows	Synthetic authentication context: <code>google_sso</code> or <code>internal</code>
<code>utm_source</code>	string	non-null	Synthetic landing attribution source; equal to <code>sessionSource</code>
<code>utm_medium</code>	string	non-null	Synthetic landing attribution medium; equal to <code>sessionMedium</code> , <code>cpc</code> for Google and <code>paid_social</code> otherwise
<code>utm_campaign</code>	string	non-null	Synthetic landing campaign name; equal to <code>campaignName</code> and <code>sessionCampaignName</code>
<code>utm_id</code>	string	non-null	Synthetic landing campaign ID; equal to <code>campaignId</code> and <code>sessionCampaignId</code>
<code>fbclid</code>	string	empty except Facebook rows	Synthetic Meta click identifier when <code>utm_source</code> is <code>facebook</code> ; never a customer key
<code>sha256_email_address</code>	string SHA-256 hex	empty for anonymous rows	Normalized synthetic email hashed as lowercase hexadecimal SHA-256 for Measurement Protocol-style user data

Column	Type	Nullability	Meaning and generation
<code>sha256_phone_number</code>	string SHA-256 hex	empty for anonymous rows	Digits-only synthetic phone normalized with +, then hashed as lowercase hexadecimal SHA-256

Simulator limitations

- `sessionSource` is not linked to a real session or user identifier.
- `sessionCampaignName` and `campaignName` are not guaranteed to correspond to their respective IDs.
- There is no `userPseudoId`, session ID, event timestamp, item array, user property, ecommerce item, or GA4 request/response envelope.
- `client_id`, `user_pseudo_id`, `user_id`, UTM fields, `fbclid`, and flattened `sha256_*` fields are simulator collection metadata; they are not all queryable as standard Google Analytics Data API dimensions.
- `totalRevenue` has no declared currency in this file.
- The `sessionMedium` implementation currently has a semantic defect documented above; consumers should not infer source-medium correctness from generated rows until fixed.

6. Zalo Official Account Metadata

Source contract

- Official documentation root: [Zalo Official Account API](#)
- Get User Detail v3.0: [Zalo Get User Detail](#)
- Endpoint represented: `/v3.0/oa/user/detail`

The current Zalo OA model is the v3.0 User Detail operation. A real response contains nested shared-information data. This dataset flattens selected profile and shared-info attributes into CSV columns.

Dataset metadata

Item	Value
File	<code>zalo_cir_api.csv</code>
Grain	One synthetic Zalo user profile
Rows	<code>Config.NUM_ROWS</code>
Primary source identifier	<code>user_id</code>
Identity fields	<code>display_name</code> , <code>shared_info_name</code> , <code>shared_info_phone</code> , <code>shared_info_city</code> , <code>shared_info_address</code>
Privacy/sensitivity fields	<code>is_sensitive</code> , <code>user_is_follower</code>
Flattening	<code>shared_info_*</code> columns represent values extracted from a real <code>shared_info</code> object

Column metadata

Column	Type	Nullability	Meaning and generation
<code>user_id</code>	string integer	non-null	Synthetic Zalo user ID, base 567826391599986760 plus row index
<code>user_id_by_app</code>	string integer	non-null	Synthetic app-scoped user ID, base 567826390000000000 plus row index
<code>display_name</code>	string	non-null	Shared synthetic person's full name
<code>user_gender</code>	integer enum-like	non-null	Shared person's gender code; code comments define 1 male, 2 female, 0 other/unknown
<code>is_sensitive</code>	boolean	non-null	Fixed <code>False</code> ; represents whether returned profile data is sensitive
<code>user_is_follower</code>	boolean	non-null	Fixed <code>True</code> ; represents follower status
<code>shared_info_name</code>	string	non-null	Flattened name from the conceptual <code>shared_info</code> object
<code>shared_info_phone</code>	string phone	non-null	Flattened phone from the conceptual <code>shared_info</code> object; CIR identity field
<code>shared_info_city</code>	string	non-null	Flattened city from the conceptual <code>shared_info</code> object
<code>shared_info_address</code>	string	non-null	Synthetic <code>Sample street N, city</code> address
<code>tags_and_notes_info</code>	string enum-like	non-null	Synthetic label from four Vietnamese-language tag values; not a documented profile identity field

Simulator limitations

- The CSV is not the raw JSON response and does not preserve the `shared_info` object.
- `tags_and_notes_info` is simulator-specific enrichment, not part of the documented user identity projection described here.
- Real Zalo permissions, access tokens, follower authorization, privacy behavior, and response error envelopes are not represented.
- Zalo documentation is rendered through a JS-heavy portal; verify exact response fields against live v3.0 response examples before treating this CSV as an integration fixture.

7. AppsFlyer Pull API Raw-Data Metadata

Source contract

- Pull API raw data: [AppsFlyer Pull API raw data](#)
- Raw data field dictionary: [AppsFlyer raw data field dictionary](#)

AppsFlyer raw-data reports use a broad, report-dependent schema. Fields can be empty or unavailable depending on report type, platform, attribution source, privacy settings, and delivery method. This simulator models an in-app-event report with a selected set of canonical Pull API field names.

Dataset metadata

Item	Value
File	<code>appsflyer_cir_api.csv</code>
Grain	One synthetic AppsFlyer in-app event
Rows	<code>Config.NUM_ROWS</code>
Identity fields	<code>customer_user_id</code> , JSON <code>event_value.customer_id</code> , device identifiers
Event set	<code>af_login</code> , <code>af_content_view</code> , <code>af_add_to_cart</code> , <code>af_purchase</code>
Time format	YYYY-MM-DD HH:MM:SS without an explicit timezone
Event value format	JSON string; purchase rows include <code>customer_id</code> and <code>order_id</code>
Revenue currency	Fixed VND for rows with synthetic revenue

Column metadata

Column	Type	Nullability	Meaning and generation
<code>attributed_touch_type</code>	string enum-like	non-null	Random <code>click</code> or <code>impression</code> ; attribution engagement type
<code>attributed_touch_time</code>	string datetime	non-null	Install time minus a random 5 to 60 minutes
<code>install_time</code>	string datetime	non-null	Base date plus random 0 to 5 days and 1 to 12 hours
<code>event_time</code>	string datetime	non-null	Install time plus random 5 to 120 minutes
<code>event_name</code>	string enum-like	non-null	Random AppsFlyer event name from the four-event set
<code>event_value</code>	string JSON	non-null	JSON containing <code>customer_id</code> ; purchase rows also contain <code>order_id</code>
<code>event_revenue</code>	decimal	non-null	Random 100,000 to 500,000 for purchase rows; 0 otherwise
<code>event_revenue_currency</code>	string currency code	non-null	Fixed VND
<code>event_source</code>	string enum-like	non-null	Fixed SDK
<code>media_source</code>	string enum-like	non-null	Random <code>facebook</code> , <code>tiktok</code> , <code>googleadwords_int</code> , or <code>zalo</code>
<code>channel</code>	string	non-null	Random display label: <code>Facebook Ads</code> , <code>TikTok Ads</code> , <code>Google Ads</code> , or <code>Zalo</code>
<code>campaign</code>	string	non-null	Synthetic campaign name; random suffix from 1 to 4
<code>campaign_id</code>	string	non-null	Synthetic AppsFlyer campaign ID <code>AF-CAMP-NNN</code>

Column	Type	Nullability	Meaning and generation
adset	string	non-null	Synthetic ad set name; random suffix from 1 to 5
adset_id	string	non-null	Synthetic ad set ID AF-AS-NNN
ad	string	non-null	Synthetic creative name; random suffix from 1 to 7
ad_id	string	non-null	Synthetic ad ID AF-AD-NNN
country_code	string ISO-like code	non-null	Shared person's country, currently VN
state	string	non-null	Shared person's state/province
city	string	non-null	Shared person's city
postal_code	string	non-null	Synthetic six-digit-looking postal code from 700010 to 700099
ip	string IPv4-like	non-null	Synthetic IPv4-like address beginning 103.21.
operator	string	non-null	Random Viettel, MobiFone, or VinaPhone; AppsFlyer field dictionary name used by current code
language	string BCP-47-like	non-null	Fixed vi
appsflyer_id	string	non-null	Synthetic SDK-generated-looking ID; random two-digit segment means uniqueness is not guaranteed
advertising_id	string UUID-like	non-null	Synthetic resettable advertising ID; row-index suffix makes this simulator-unique
idfa	string	empty on Android	Synthetic iOS advertising identifier; empty when is_android is true
android_id	string	empty on iOS	Synthetic Android identifier; empty when is_android is false
customer_user_id	string	non-null	Shared synthetic cid, strongest AppsFlyer-to-CIR key
idfv	string	empty on Android	Synthetic iOS vendor identifier; empty when is_android is true
platform	string enum-like	non-null	android or ios , selected by is_android
device_type	string	non-null	Synthetic device model label; AppsFlyer field dictionary notes this field is deprecated/unpopulated in favor of device model in current reports
os_version	string	non-null	Random Android 14/15 or iOS 17.6/16.7
app_version	string	non-null	Fixed 6.8.1
sdk_version	string	non-null	Fixed 6.15.0
app_id	string	non-null	Fixed synthetic app ID com.example.customer360

Column	Type	Nullability	Meaning and generation
<code>bundle_id</code>	string	non-null	Fixed synthetic bundle/app ID <code>com.example.customer360</code>
<code>user_agent</code>	string	non-null	Synthetic Android or iOS browser-style user agent
<code>http_referrer</code>	string URL	non-null	Fixed synthetic landing URL
<code>original_url</code>	string URL	non-null	Synthetic app URL containing the person's <code>cid</code>

Simulator limitations

- This is a selected subset of the AppsFlyer main schema, not a complete raw-data field dictionary export.
- The timestamp strings omit timezone information even though delivery method and account settings affect real AppsFlyer timestamp timezone behavior.
- Attribution hierarchy values are independently randomized and are not a stable campaign entity model.
- `appsflyer_id` can collide because only a limited random suffix is used; do not use it as a guaranteed unique key in tests.
- Device identifiers are synthetic and must not be interpreted as real device identity.

8. Cross-Platform CIR Metadata

Identity key availability

Platform	Strongest shared key	Secondary keys	Direct <code>cid</code> column?
Meta	<code>field_email</code> , <code>field_phone</code>	First/last name, city, state	No
TikTok	None	Advertiser/campaign hierarchy and country only	No
GA4	None in current schema; production GA4 may use non-PII <code>user_id</code>	City, country, campaign/source context, optional UTM-derived traffic dimensions	No
Zalo	<code>shared_info_phone</code>	Name, city, address	No
AppsFlyer	<code>customer_user_id</code> , <code>event_value.customer_id</code>	Original URL, device identifiers, geography	Yes, embedded in fields rather than as a column

Expected synthetic row-to-person mapping

- Meta, GA4, Zalo, and AppsFlyer select shared people cyclically.
- With 20 rows and 8 people, person positions repeat after every eight rows.
- TikTok rows are aggregate advertising observations and intentionally have no person-level identity.
- A real CIR implementation should normalize email and phone before matching and should treat names, city, and device identifiers as supporting evidence rather than guaranteed unique identifiers.

Data quality and reproducibility

- The module seeds Python's global random generator with 42 at import time.
- Importing this module has side effects: it creates the configured output directory and changes the process-global random sequence.
- Reproducibility assumes the same Python implementation, execution order, configuration, and generator code.
- CSV values are serialized from Python values; downstream ingestion should explicitly cast numeric, boolean, date, datetime, and JSON-string fields instead of relying on inference.

9. Source References and Maintenance Rules

Update this document when any generated header, source endpoint, field meaning, type, format, identity behavior, or flattening rule changes.

Before using a generated file as an integration fixture:

1. Run the simulator from the repository root.
2. Confirm every CSV header against the tables in this document.
3. Parse `event_value` as JSON for AppsFlyer rows.
4. Validate GA4 date and metric conventions against the current Data API schema.
5. Validate TikTok dimensions and metrics against the report type and data level being simulated.
6. Validate Meta and Zalo flattened fields against the corresponding live object examples.
7. Treat all IDs, names, revenue, and device values as synthetic test data.

10. Pandoc PDF Export

From the repository root, export this document with XeLaTeX:

```
pandoc ./all-data-simulator/full_metadata_doc_simulator.md --pdf-  
engine=xelatex -o ./all-data-simulator/full_metadata_doc_simulator.pdf
```

The YAML metadata in this file enables the table of contents, A4 page geometry, wrapped URLs, compact long tables, and clickable PDF links. The export requires Pandoc and a XeLaTeX installation with the `microtype`, `xurl`, and `enumitem` packages available.